

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Winter Examinations 2023**

**Course: B. Tech. Branch : Computer Science & Engineering Semester :VI**

**Subject Code & Name: Computer Network (BTCOC602)**

**Max Marks: 60**

**Date:**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. All Questions are Compulsory.
2. Draw neat diagram wherever necessary.
3. Figures to right indicates full marks
4. Assume suitable data wherever necessary and mention it clearly

(Level/CO) Marks

**Q. 1 Solve Any Two of the following.**

- |                                                                  |              |           |
|------------------------------------------------------------------|--------------|-----------|
| A) Describe social issues with computer network?                 | (Understand) | <b>06</b> |
| B) Give comparison between OSI reference model and TCP/IP model? | (Understand) | <b>06</b> |
| C) Compare connectionless and connection oriented protocol?      | (Understand) | <b>06</b> |

**Q.2 Solve Any Two of the following.**

- |                                                                                          |              |           |
|------------------------------------------------------------------------------------------|--------------|-----------|
| A) Discuss Wi-MAX (802.16) and Bluetooth (802.15.1) technology application in real life? | (Understand) | <b>06</b> |
| B) Compare token ring and FDDI with their frame format?                                  | (Understand) | <b>06</b> |
| C) Describe in brief 802.11 architecture and protocol stack?                             | (Understand) | <b>06</b> |

**Q. 3 Solve Any Two of the following.**

- |                                                                                                                                                        |           |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| A) Illustrate byte count and bit stuffing framing method?                                                                                              | (Apply)   | <b>06</b> |
| B) A (7,4) hamming code is received as 1110000. Determine the correct code when even parity is considered.?                                            | (Apply)   | <b>06</b> |
| C) Justify how CRC is a stronger kind of error-detecting code.<br>msg – 1 1 0 1 1 0 1 0 0 0 0 gen- 1 0 1 0 1 , Calculate CRC for the given data input? | (Analyze) | <b>06</b> |

**Q.4 Solve Any Two of the following.**

- |                                                                                                                               |              |           |
|-------------------------------------------------------------------------------------------------------------------------------|--------------|-----------|
| A) Compare IPv4 and IPv6 address formats?                                                                                     | (Understand) | <b>06</b> |
| B) Describe the following congestion control mechanisms with suitable example.<br>1. Traffic Throttling      2. Load Shedding | (Understand) | <b>06</b> |
| C) Illustrate with example leaky bucket and token bucket algorithms for traffic shaping?                                      | (Understand) | <b>06</b> |

**Q. 5 Solve Any Two of the following.**

- |                                                    |              |           |
|----------------------------------------------------|--------------|-----------|
| A) Describe five tuple DNS record with an example? | (Understand) | <b>06</b> |
| B) Compare SMTP and POP Protocols?                 | (Understand) | <b>06</b> |

- C) Illustrate the network security essentials Confidentiality, Authentication, Non repudiation and Integrity control for considering the E-commerce website?

(Apply)

**06**

**\*\*\* End \*\*\***